A. Next Test

time limit per test: 2 seconds
memory limit per test: 256 megabytes

«Polygon» is a system which allows to create programming tasks in a simple and professional way. When you add a test to the problem, the corresponding form asks you for the test index. As in most cases it is clear which index the next test will have, the system suggests the default value of the index. It is calculated as the smallest positive integer which is not used as an index for some previously added test.

You are to implement this feature. Create a program which determines the default index of the next test, given the indexes of the previously added tests.

Input

The first line contains one integer n $(1 \le n \le 3000)$ — the amount of previously added tests. The second line contains n distinct integers $a_1, a_2, \ldots, a_n$ $(1 \le a_i \le 3000)$ — indexes of these tests.

Output

Output the required default value for the next test index.

Examples

input Copy

```
3
1 7 2
```

output Copy

```
3
```

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Beta Round 27 (Codeforces format, Div. 2)

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, msys2

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
372416428	Apr/25/2026 08:44	Accepted
372416026	Apr/25/2026 08:39	Accepted
372415949	Apr/25/2026 08:37	Compilation error

→ Problem tags

implementation sortings *1200

No tag edit access

→ Contest materials

- Announcement
- Tutorial #1
- Tutorial #2 (en)
- Tutorial #3 (en)

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The only programming contests Web 2.0 platform
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